



current is not high enough to switch on a neon lamp.

This simple circuit can detect the electric current passing through earth wires. It is a remote-type tester and direct contact is not necessary to check the presence of current. It can also be used to check the presence of current in power cables, switches, etc.

The circuit picks up the electromagnetic field from the AC line through a coil sensor. Here the sensor used is the coil of an ordinary 6V relay. The electromagnetic field from the AC generates a voltage in the coil through induction. The AC thus generated is rectified by the bridge rectifier comprising 1N4007 diodes (D1 through D4). The rectified DC then passes to the non-inverting input (pin 3) of IC1. The high input impedance makes IC1 very sensitive to minute

voltage variations.

The open-loop voltage gain of TLO71 is around 100 dB. The inverting input of IC1 receives an adjustable voltage through preset VR1. Resistor R1 protects the inverting input if the wiper of preset VR1 is fully turned. R2 is the feedback resistor.

The output of IC1 is given through VR2 to the signal input (pin 5) of IC2. Resistor R3 also acts as a protective resistor. Monolithic IC LM3915 (IC2) is used to get an analogue display. The output of IC2 goes low one by one with an increment of 0.5 volt in input pin 5. Pin 9 of IC2 is left unconnected to get a dot-mode indication. LED1 lights up when input pin 5 receives 0.5 volt and LED2 lights up when the voltage increases to 0.75 volt. LED3 lights up when the voltage increases to 1 volt. Thus it is easy to detect the strength

of the electric current through the LED display. Resistor R5 controls the brightness of the LEDs.

Calibration and setting require some trial and error. Keep the wiper of preset VR2 in the extreme anti-clockwise position and slowly adjust VR1 until LED1 lights up. This indicates normal electromagnetic

field around the circuit. Take the probe near the AC line. LED2 will light up to indicate the presence of electric current. When the distance becomes shorter, LED3 lights up and LED2 turns off.

After setting, glue the wiper of VR1 and VR2 with adhesive. Take the probe near the earth-grounding pipe. If LED2 and LED3 light up, it indicates the drainage of current through leakage.

Assemble the circuit on a general-purpose PCB and enclose in a plastic case. Use a probe similar to that of the multimeter. Connect the probe to the circuit using a 30cm long flexible wire.

EFY note. Do not touch the probe to bare electric points or AC sockets for testing. It is not necessary to touch the probe and a mere presence of electric current is sufficient for the circuit to get activated and be functional. ●

SIXTH-ORDER BUTTERWORTH LOW-PASS FILTER

■ VASUKI PRASAD

Low-pass filters are widely used in the audio domain. Their most important application is in communication receivers and communication audio circuitry. These filters, when used for audio communications, start attenuating frequencies above 3.4 kHz, as the bandwidth of the human voice

(as far as voice clarity and intelligibility are concerned) is 300 Hz to 3.4 kHz.

Here we describe a low-pass filter that exhibits a flat passband and attenuation rate of 120 dB per decade after 3.4 kHz. This means that a signal with a frequency of 34 kHz is attenuated by 120 dB compared to a signal of 3.4 kHz. No ripples are seen in the passband and stopband. A filter with

flat passband and ripple-free attenuation band is known as a 'Butterworth' filter.

Although the Butterworth circuitry is supposed to exhibit a gain of 1 (or 0 dB), a small gain is seen in the circuit. Nevertheless, the shape of the transfer curve (frequency domain) is unaffected, and this minor gain is in no way detrimental to the performance

